

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

Nov 2012

CL 208 Advance Surveying

Date: 05.11.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Derive distance and elevation formula for horizontal line of sight having fixed hair stadia. [04]

(b) Explain the term systematic error and accidental error. [03]

Q - 2 (a) A tacheometer is setup at point P [05]

Staff Station	Vertical angle	Hair readings	Remarks
Q	$+6^{\circ} 36'$	1.905, 2.480, 3.055	R.L of BM 852.55
BM	$+0^{\circ} 00'$	1.855	

The instrument constant are 100 & 0.5 .Compute the length PQ and the reduce level of Q.

(b) Adjust the angle P, Q, R and S which close the horizon. [05]

$\angle P = 115^{\circ} 32' 18.8''$ Weight 3 $\angle Q = 122^{\circ} 46' 21.4''$ Weight 4

$\angle R = 62^{\circ} 21' 16.8''$ Weight 2 $\angle S = 59^{\circ} 20' 15.8''$ Weight 1

(c) Explain key components of GIS. [04]

OR

Q - 2 (a) Derive elevation and distance formula for tangential method, when both angles are those of elevation. [05]

(b) Angles were measure on a station and the observations were recorded as follows. [05]

$\angle A = 70^{\circ} 31' 18.6''$ Weight 3 $\angle B = 61^{\circ} 12' 9.8''$ Weight 2

$\angle C = 112^{\circ} 41' 13.6''$ Weight 4 $\angle A+B = 131^{\circ} 43' 20.6''$ Weight 2

Find the most probable vale of angle A,B and C.

(c) What do you understand by GPS? Write application of GPS. [04]

Q - 3 (a) Derive elevation and distance formula for Inclined sight with staff held vertical, angle [04]

of elevation.

- (b) Define [04]
(i) Observed Value (ii) True Value (iii) Most Probable value (iv) Weight
(c) Explain active and passive system of remote sensing. [06]

OR

- Q - 3 (a) Derive elevation and distance formula for inclined sight with staff held normal, angle of elevation. [04]
(b) Explain Accidental errors in detail. [04]
(c) Write applications of remote sensing. Explain any one in detail [06]

SECTION - II

- Q - 4 (a) Define following [05]
(i) Altitude (ii) Picture Plane (iii) Ground Plane (iv) Principal point (v) Floating marks
(b) (i) Convert the time into degree 4h 12m 48s [02]
(ii) Convert the degree into time $12^{\circ} 32' 48''$

- Q - 5 (a) Determine the R.L. of the top of a hill from the following observations. [07]

Inst. Station	Vertical angle top of hill	Axial hair reading m	RL. of A m
A	$9^{\circ} 30'$	2.0	428.650
B	$12^{\circ} 30'$		

Instrument axis at A and B at the same level, point B is near to hill. The distance AB=1600m. Determine elevation of top of the hill and its distance from point A.

- (b) Explain overlaps and number of photographs in context of aerial surveying. [07]

OR

- Q - 5 (a) To determine height of chimney, a theodolite was kept at two station I_1 and I_2 200m apart, I_1 being nearer to chimney. The readings at the bench mark (of RL 1020.375) were 1.35m from station I_1 and 2.15m from I_2 . The vertical angles to the top of the chimney were $19^{\circ} 30'$ and $8^{\circ} 15'$ from stations I_1 and I_2 respectively. Find the horizontal distance and RL of the top of the chimney [07]

- (b) Define Parallax and derive parallax equation. [07]

- Q - 6 (a) What do you understand about triangulation? Explain principal of triangulation. [07]

- (b) Define and show following in a sketch- horizon, declination, hour angle, meridian, altitude, right ascension. [07]

OR

Q - 6 (a) Explain various triangulation systems.

[07]

(b) Calculate the hour angle and azimuth of the sun from the following data.

[07]

(a) Latitude of the place is 46° N, declination is 22° S at sun rise

(b) Latitude is 46° S declination is 20° N at sunset.
